NEOGEOSDK / V1.7.7 · EAGLE SOFTWARE · MIT

NEO·GEO SDK

The games release

Seven games built to real ROMs, four genre engines, a fighting core, an open system ROM and an experimental Neo Geo CD backend, on direct 68000 control of video, palettes, sprites, FIX tiles and a Z80-driven YM2610 sound stack.

INSERT COIN

SNK ARCADE / HOME · 68000 + Z80 + YM2610 · MVS · AES · NEO GEO CD (EXPERIMENTAL)

THE PITCH

Real Neo Geo development, made practical.

For self-made games, demos and homebrew that stay close to how the machine was programmed, with modern build tooling, scripting and emulator-based testing around it.

Target	SNK Neo Geo: MVS (arcade), AES (home); Neo Geo CD experimental
Release	v1.7.7 · MIT licensed · SDK 1.7.7, Framework 1.5
Host platforms	Linux · WSL · Windows
Languages	C99 · C++14 · 68000 asm · Z80 asm · Python
Audio	FM · SSG · ADPCM-A · ADPCM-B
Games	Seven, each with MVS and AES sets, NeoSD, Darksoft and BackBit packages
Engines	Plain C and C++14, one ABI; opt-in shooter, fighting and genre cores

AT A GLANCE

Five pre-releases. One release.

V1.7.1

Framework 1.0

The C engine frozen and versioned; video written in the blank; sound commands queued. Maiya and EagleBIOS.

V1.7.2

Frames in the blank

A deadline, priorities, moves first: no frame ever written on a drawn line.

V1.7.3

Shooters

ng_shooter, pinned character slots. Sky Lance's first set.

V1.7.5

Fighting

ng_fight: motions, cancels, teams, strikers. Shadow League. EagleBIOS v1.7.4.

V1.7.7

Genre engines

Puzzle, mahjong, RPG, beat'em-up; Neo Geo CD; the ROM registry; seven games released.

COMPATIBILITY

Every release since Framework 1.0 adds and never changes: a game written for v1.7.0 builds unchanged on v1.7.7.

THE TARGET MACHINE

Three chips. One bus. No engine.

MAIN CPU

M68000 @ 12 MHz

Game logic, video memory, palettes, sprites, FIX tiles, directly.

SOUND CPU

Z80A @ 4 MHz

Runs the SDK's driver; takes commands from the 68000 through ports.

SOUND CHIP

YM2610

FM synthesis, SSG square waves, ADPCM-A effects and ADPCM-B music.

THE DEVELOPMENT MODEL

Graphics, palettes, FIX tiles, ROM layout and sound commands are part of everyday programming rather than hidden behind a large engine. The SDK keeps that: it gives you headers, build glue, asset pipelines and a 2D engine you opt into, not abstractions you cannot see past.

01

The games

Seven games build to real ROMs from this repository, each for MVS and AES.

WHAT BUILDS FROM THIS REPOSITORY

The shipped games.

GAME	ID	GENRE	IN IT
Maiya: Super Nature Girl	780	Action platformer	Twelve stages, guardians behind sealed gates, hidden vaults; Maiya or her sister Luna
Sky Lance	779	Vertical shooter	Seven painted routes to boss arenas, three jets
Shadow League Fighting	781	Fighting	44 fighters, 12 arenas, team battles with strikers, voices, score and bonus counts
No Signal Z Alpha	784	Beat'em-up	Eight streets for one player, grabs, throws, a scooter
Maiya & Luna	783	Adventure RPG	Two solo journeys, seven chapters, puzzles, saves
Seven Wonders Puzzles	782	Puzzle	Match, bubble and picture-logic boards, an album
Mahjong Demo	785	Mahjong	Hong Kong Old Style against three CPU players, solitaire

SHADOW LEAGUE FIGHTING · 781

A fighter, front to back.

- **44 painted fighters, 12 arenas**: single matches and three-on-three team battles with a striker per team, a referee who calls the round, voices for every move.
- **Every round ends like one**: the winner's pose over the loser, team bouts included; the beaten team kneels at the match's end.
- **A score with bonus counts**: life, time and untouched bonuses after each round won, counted into the score; TRIPLE UNTOUCHED and CLEAN SWEEP finales with their own fanfares.
- **Arenas in 32 palettes**, fitted tile by tile: skies, beams and wet floors keep their gradients.
- **Combos for everyone**: a connected normal cancels into any special or super, a special that hits into a super, beside each fighter's own routes.

ENGINES IN USE

Each game shows an engine.

MAIYA · 780

Platformer

Deferred video, pinned hero slots, raster bands, landmarks loaded at a stage's start; busy fights at full speed.

SKY LANCE · 779

ng_shooter

Scroll, camera, parallax, a strip budget per line; fixed-point aimed shots, anchored clouds.

NO SIGNAL · 784

ng_beatemup

Waves, a crowd director, hitstop per move, stride-driven walk and run, ridable vehicles.

MAIYA & LUNA · 783

ng_rpg

Rooms, people, items and environmental puzzles from ROM tables; saves.

SEVEN WONDERS · 782

ng_puzzle

Match, bubble and picture-logic boards with rewards.

MAHJONG DEMO · 785

ng_mahjong

A four-seat table against the CPU, and solitaire.

02

The engine

One engine in plain C or C++14; opt-in cores for shooters, fighters and four genres.

ADDITIONS ONLY

The engine only grows.

FRAMEWORK	RELEASE	ADDED
1.0	v1.7.1	Public calls frozen (docs/api/framework_v1.txt, make api-check); VRAM in the blank; sound queue
1.1	v1.7.2	Raster bands on the line timer (ng_raster)
1.2	v1.7.2	VRAM priorities and the blank's deadline
1.3	v1.7.3	ng_shooter; pinned character slots; depth-sorted cast
1.4	v1.7.5	ng_fight, the fighting core
1.5	v1.7.7	Genre engines; 20-bit tile addresses; frame sync; move-first groups

NG_VRAM_DEFER

Nothing is written on a drawn line.

DEADLINE

Line 8

No job starts after it: what does not fit waits for the next blank.

ORDER

Moves first

Then content by priority: HIGH, streaming strips, NORMAL, FIX text, LOW.

MOVE-FIRST GROUPS

NG_SG_MOVE_FIRST

A group already showing its picture keeps moving with the scenery while its new picture waits.

FRAME SYNC

One batch a frame

The main loop publishes a finished frame; the interrupt takes it whole.

WHY IT MATTERS

A frame that tears or shows half a picture never reaches the screen: it waits a frame instead, and the game measures how often that happens (make PERF=1).

ONE LINE IN GAME.MK

Six cores, picked per game.

-DNG_SHOOTER=1

Shooter

Vertical scroll, smooth follow, paced parallax, strip budget.

-DNG_FIGHT=1

Fighting

Motions, cancels, super freeze, projectiles, teams, strikers, CPU ranks, training.

GAME_GENRE_ENGINE = PUZZLE

Puzzle

Match and bubble boards, picture logic, hints.

= MAHJONG

Mahjong

Hong Kong Old Style for four seats; solitaire layouts.

= RPG

RPG

Rooms, exits, party, items, dialogue, environmental puzzles, saves.

= BEATEMUP

Beat'em-up

Waves, crowd director, grabs and throws, pick-ups, vehicles.

THE FIGHTING CORE

Fighters are data.

- **Input**: a 64-frame ring, quarter and half circles, dragon punch, charge, 360 and 720, negative edge, two buttons a frame apart still read together.
- **Moves**: authored cancel routes per fighter plus shared ones; juggle limits; MAX mode; guard cancels; throws with grips, carries and breaks.
- **Matches**: rounds or three-on-three teams with strikers; the game can hold a round's end for its own ceremony; a deterministic match hash.
- **CPU and training**: seven calibrated ranks; recording and playback dummies.

MEASURED IN MAME, FRAME BY FRAME

16.7 ms. That's the budget.

COLOUR MIXING

45 divides to 0

A 16-colour tint used the library's 32-bit divide 45 times; shifts and adds now give the same colours.

MAIYA, WORST STAGE

50 to 57 fps

Golden Savanna's fights, before and after the release's performance pass.

DROPPED FRAMES

17.5% to 3.1%

Same stage, same run, the share of frames that overran.

RULE 14: DIVIDE IN 16 BITS

The 68000 divides 32 bits by 16. A division the compiler cannot prove fits becomes a library call several times slower; docs/performance_rules.md lists the rules every game follows.

03

The platform

EagleBIOS for MVS and AES, and an experimental road to the Neo Geo CD.

THE OPEN SYSTEM ROM

Our own BIOS, in every package.

- **MVS and AES**: boots every SDK game; packages ship EagleBIOS and no other BIOS.
- **Memory cards**: 16-bit JEIDA/PCMCIA detection, a save directory, errors; a checksummed backup RAM format with migration.
- **Clock and menus**: the real-time clock, settings, cabinet and game DIPs, diagnostics, sound and input tests, cartridge information.
- **HOW TO PLAY**: a game's own text page (HTP1), with joystick and button badges.

EXPERIMENTAL

From cartridge to disc.

PROGRAMS

PRG and IPL

Native CD programs from the same C sources; format and bank checks.

DISCS

ISO and BIN/CUE

Authored by the SDK; one BIN/CUE per disc for SD loaders and burning.

PORTS

Four games

Maiya, Sky Lance, Maiya & Luna and No Signal, with streamed scenery and CD audio.

STATUS

All four discs boot in MAME with stock CDZ firmware and with EagleBIOS-CD. Saves under EagleBIOS-CD and physical discs are still to be verified: see docs/NEOGEO_CD_PIPELINE_AND_TESTING_GUIDE.md.

04

Shipping

Versions that mean one set of bytes, packages for every machine, tests on both.

RELEASES/ROM-REGISTRY.JSON

One name, one set of bytes.

- **A version per game**, MAJOR.MINOR.PATCH: new content, improvements, fixes. Changed bytes always get a new version.
- **A build identity**: twelve hex digits over the hashes of the six ROM parts; a MAME set, a Darksoft folder and a BackBit zip of one build share it.
- **Audits**: make_release.py refuses to replace a version that went out; audit_release.py checks every package; rom_registry.py --verify checks the release folder.

MAME, BOTH MACHINES

Tried in MAME, every build.

MAME_BATS.PY

Launchers

A menu, every game for MVS and AES, level launchers that start any stage, video recording steps.

INSTALL_WIP_MAME.PY

Installs

A build into the test folder with its own software list, settings and saves.

RELEASE_ALL_GAMES.SH

Pipeline

Every game built, host tests, registry checks, boots, a QA summary.

GAME QA

Scripts

Matches played, captures at round ends, frame and VRAM counters, instruction profiles.

FOR EACH GAME, EACH VERSION

Emulator, flash cart, cabinet.

PACKAGE	FOR
-WIP-EagleBIOS.zip / -WIP-AES-EagleBIOS.zip	MAME: the set, its software list, EagleBIOS (AES: a blank memory card)
-NeoSD_MVS.neo / -NeoSD_AES.neo	NeoSD: cartridge data, the system's own BIOS
-Darksoft-MVS.zip / -AES.zip	Darksoft Neo Geo Multi
-BackBit-MVS.zip / -AES.zip	BackBit Platinum
-README / -CHANGELOG / -MANIFEST	How to play; what changed; sizes, CRC-32 and SHA-256

FROM A CLEAN CHECKOUT

make GAME=<name> all make test

```
# Linux / WSL
$ make GAME=shadowleague all
$ make neo GAME=maiya
$ tools/release_all_games.sh

# Windows
$ make -f MakefileWin32.mak GAME=skylance all
```

REPOSITORY

github.com/eaglesoftware777/neogeosdk

Source, issues and releases.

RELEASE

v1.7.7

SDK 1.7.7 · Framework 1.5

LICENCE

MIT

Use it, ship games with it.